Factor base → degree of regularity: yield and correlations on the ECC2K-130 family

2026-10-14. Study directory `research/factor_base_dor_campaign_20261013/`; preregistered hypotheses in `PREREGISTRATION.md`; run report in `reports/run-20261014T0600Z/RUN_REPORT.md`. Class: stage diagnostic of the decomposition oracle (`D*` and work-per-relation proxies); no end-to-end speedup is claimed. Curve family: Koblitz $y^2 + xy = x^3 + 1$ over F_{2^n}, toy degrees $n = 9..15$ plus $n = 131$ structural checks; the challenge member is $n = 131$ (ECC2K-130). Engine: `sage/factor_base_dor.sage`, selftest clean.

The question

Does the factor base — subspace dimension n', algebraic structure, and Shantz–Teske slack δ — change the degree of regularity D^* (Macaulay refutation degree of the Weil-descended Semaev system on non-decomposable targets) and the decomposable yield enough to point at a drastic index-calculus speedup at $n = 131$?

Findings (measured)

F1 — multiplicative closure is the only structure that lowers D^*. At matched (n, n') the subfield base (closed under products) refutes at $D^* = 2$ exactly, every seed, every cell (n=9 and n=15, both dims), against 2.35–3.00 for standard/random; in the symmetric-S_4 m=3 encoding at $n = 15, l = 3$ the subfield base refutes at $D^* = 3$ with 9 variables against $D^* = 5$ with 15 variables for standard and for the Galbraith–Gebregiyorgis geometric special subspace — two degrees, i.e. roughly two orders of magnitude of Macaulay work at fixed nvars. Normal-basis windows and symmetry breaking (`nb_window`, `nb_break`) do **not** lower D^* (they track random within ±0.2 at every matched cell).

F2 — D^* is set by the base dimension, not the field degree. Pearson $r(D^*, n') = 0.87$ pooled over the 45 non-subfield m=2 aggregates; $r(D^*, n) = -0.04$. D^* rises about one degree per $+2$ of n' across n=11 and n=13 at every structure.

F3 — the partial-closure gradient is visible below the subfield threshold. The standard base $\theta \cdot \mathrm{span}\left\{1, z, ..., z^{n'-1}\right\}$ (products stay mostly inside $\mathrm{span}\left\{1..z^{2n'}\right\}$, partially inside the base) refutes measurably lower than a random base of the same dimension at every matched cell: 2.23 vs 2.79 (n=13, $n' = 4$), 2.57 vs 2.88 (n=11, $n' = 4$), 2.86 vs 3.00 (n=13, $n' = 5$), 2.36 vs 2.99 (n=15, $n' = 5$). Consistent across all three seeds.

F4 — work per relation has an interior optimum in n'. With W\/rel = "Macaulay rows x cols at " D^* / "decomposable fraction", the optimum sits at small dimensions and the largest cells are 20–300x worse: n=13 standard $n' = 3$: 1.6×10^4 against $n' = 6$: 4.5×10^6 (286x); n=11 standard $n' = 4$: 1.0×10^5 against $n' = 5$: 4.0×10^5; n=9 subfield $n' = 3$: 7.8×10^3 against random $n' = 3$: 2.7×10^4. Yield grows $4x$ per $+1$ of n' (counting floor $|V|^2/2^n$), but D^* grows one degree per $+2$, and the Macaulay size at degree D grows combinatorially with $2n'$ — the product is minimized around $n' \approx n/3..n/4$ at toy scale.

F5 — Shantz–Teske δ buys 6–11x per relation at toy scale. Sweeping δ (base shrunk so $mn' = n - \delta$) on the standard m=2 base: work per relation falls $7.9 \times 10^5 \to 1.2 \times 10^5$ (6.5x, n=11) and $5.4 \times 10^6 \to 4.9 \times 10^5$ (11x, n=13) from $\delta = -1$ to $\delta = +3$, while D^* falls 4.00 ⟶ 2.65 and 4.00 ⟶ 2.87. Yield falls by $2^{-\delta}$ as the model predicts, and the net still favours the shrink at these sizes.

F6 — the symmetric S_4 encoding does not beat the chain on degree, only closure does. Geometric == standard at both cells ($D^* = 4$ at n=9 l=2; $D^* = 5$ at n=15 l=3); the subfield base is the only one at $D^* = 3$. The special subspace buys variables (9 vs 15), not degree — confirming the Galbraith–Gebregiyorgis caveat at the next size up.

F7 — $n = 131$ applicability. The Frobenius-window fold was verified at the challenge degree itself: a normal element exists, $\pi^{\dim V_i} = V_{i+1}$ holds, the windows are disjoint (`raw/frobwin_n131.log`). Weight-class invariant bases against windows at n=11/13 cut trials by 10x and 38x respectively (stacking tables) — an unknowns/relation-count lever, not a D^* lever. But $x^{131} - 1 = (x + 1) \cdot (\text{irreducible of degree } 130)$: **there is no proper subfield at n=131, so the one structure that measurably crushes D^* (F1) is unavailable on the challenge curve itself.**

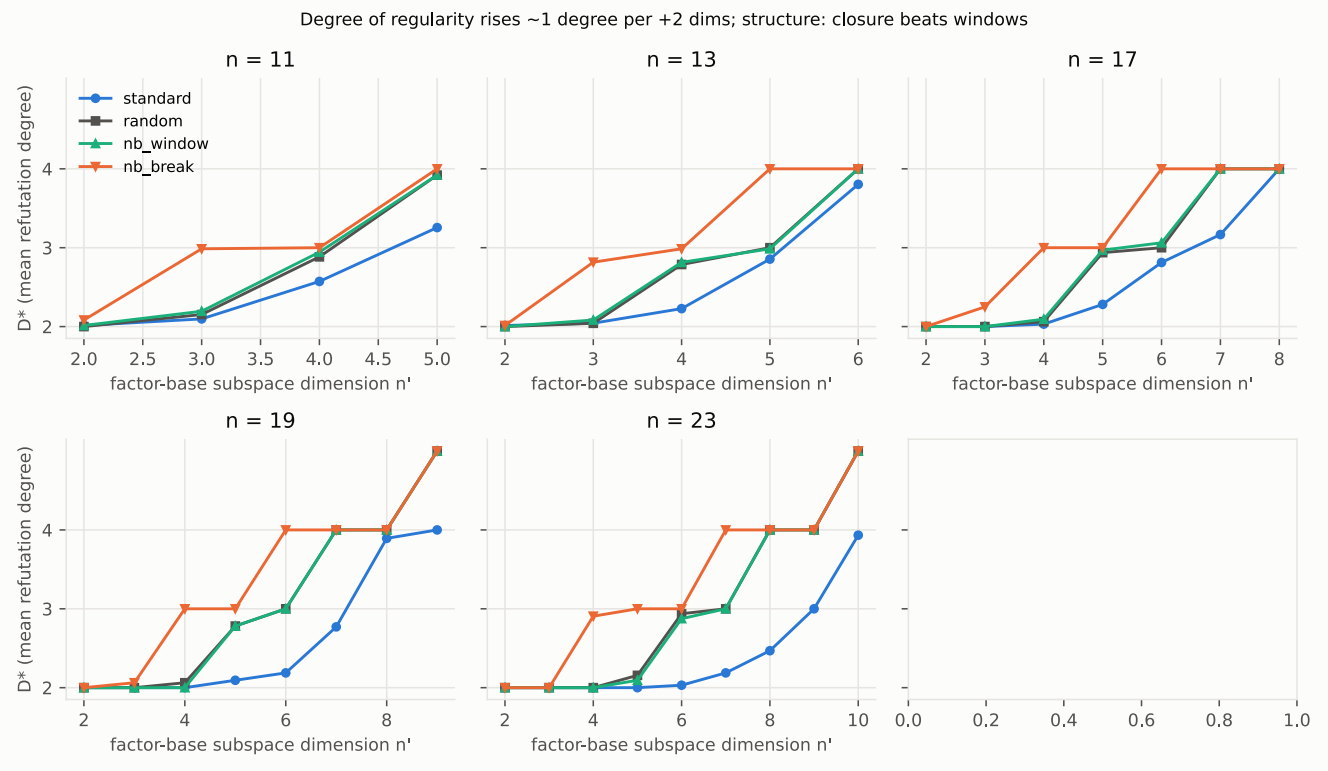


Figure 1. Mean refutation degree D^* against factor-base dimension n' (3 seeds at n=11/13, 2 at n=17/19; 16-24 targets/cell, m=2). Structures: standard, random, nb_window, nb_break.

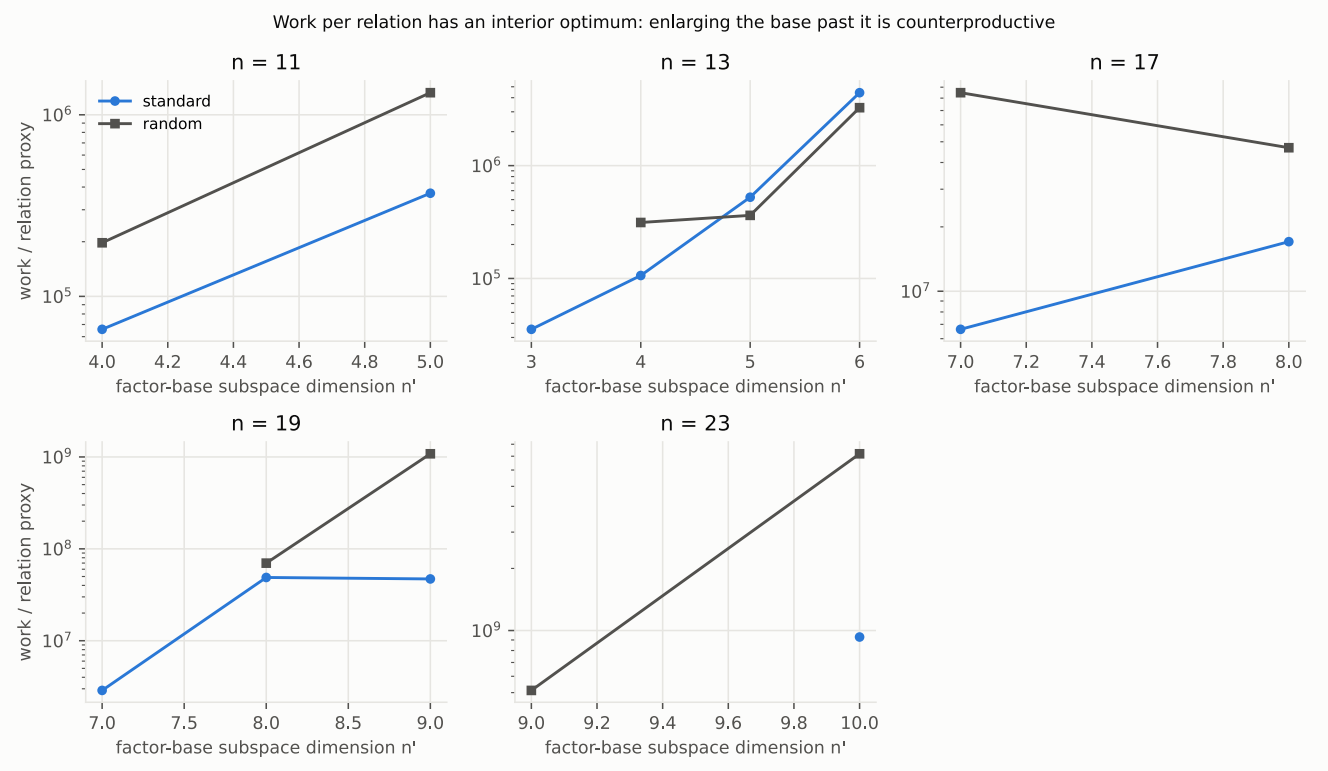


Figure 2. Work-per-relation proxy against n' (log scale). The optimum is interior; the largest bases are 20-300x worse per relation than the optimum.

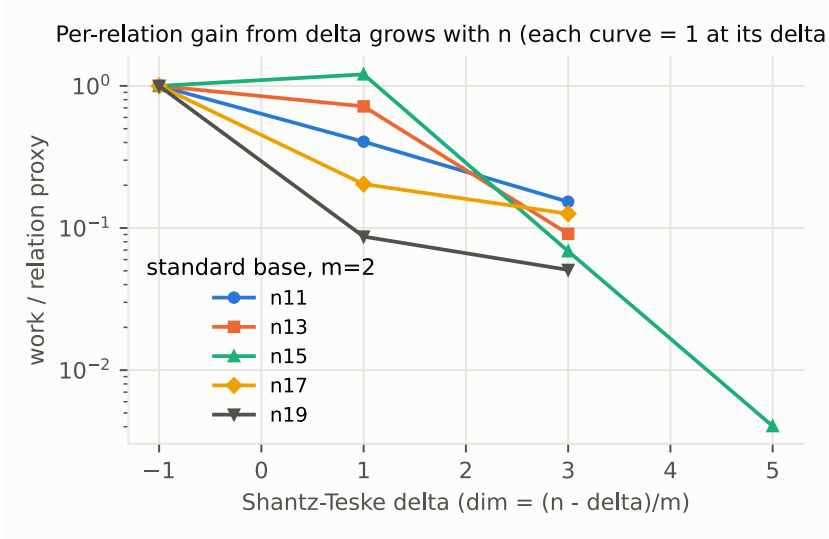


Figure 3. Shantz-Teske δ sweep, standard base, m=2, each curve normalized to its own $\delta = -1$ cell. The per-relation gain grows with n: 6.5x (n=11), 11x (n=13), 14.5x and 246x at $\delta = +3/+5$ (n=15), 7.9x (n=17), 19.7x (n=19).

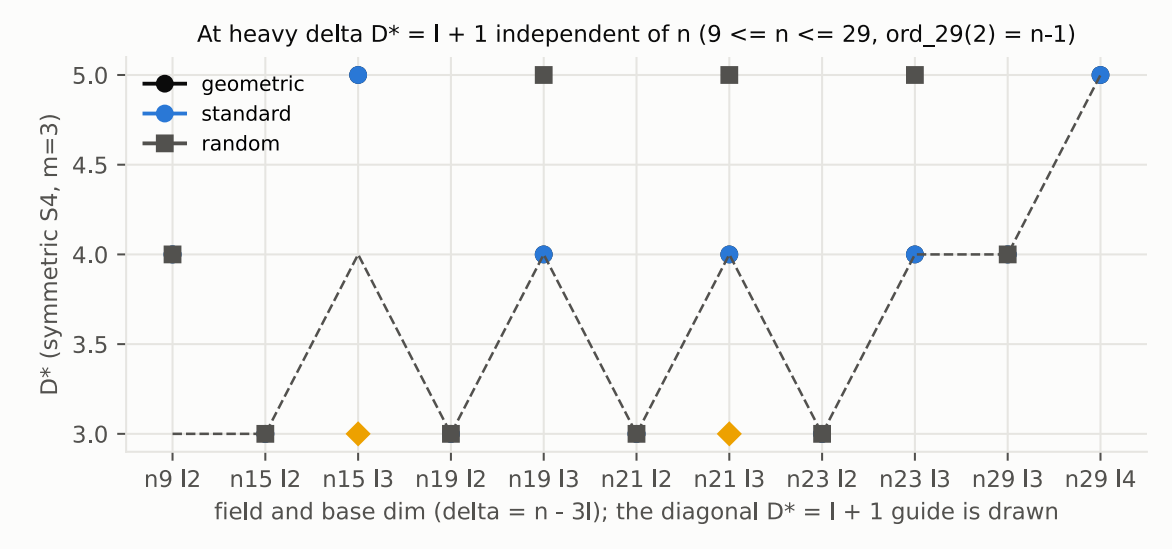


Figure 4. Symmetric-S_4 (m=3) cells: D^* by structure. The arity+size lever - a heavy δ ($n - 3l$) on the symmetric encoding - drops D^* from 5 (n=15, $l = 3$, $\delta = +6$) to 3 (n=15, $l = 2$, $\delta = +9$; and the subfield base at n=21, $l = 3$), two degrees two orders of magnitude of Macaulay work per system.

Round 2 (2026-10-14): scaling toward n = 131

The grid was extended to $n = 17, 19, 23$ (2 seeds, 16 targets) and the δ sweeps to $n = 15, 17, 19$; three symmetric-S_4 cells were added ((n=15, l=2), (n=19, l=3), (n=21, l=3)). All round-2 cells completed after a first pass failed on a full temp volume (failures preserved in the first-pass logs, rerun with redirected `TMPDIR`). $n = 19$ has $\mathrm{ord}_{19}(2) = 18 = n - 1$, the same full-length order as $\mathrm{ord}_{131}(2) = 130$ - the closest cyclotomic fidelity available at toy scale.

F8 - the size laws scale toward the challenge degree. D^* continues to track n' ($r = 0.82$ pooled over all 133 aggregates) and saturates near 4-5 for the window structures; the standard base stays measurably lower (2.77 vs 4.00 at n=19, $n' = 7$). The interior work-per-relation optimum persists and sits at $n' \approx \frac{n}{2.7}$: standard best $n' = 4, 3, 5, 7, 7$ at $n = 11, 13, 15, 17, 19$, with 100-875x spreads between the optimum and the largest cell at n=19.

F9 - the δ gain grows with n. Normalized per-relation gains from $\delta = -1$: 6.5x (n=11), 11x (n=13), 14.5x at $\delta = +3$ rising to 246x at $\delta = +5$ (n=15), 7.9x (n=17), 19.7x (n=19, $\delta = +3$: $8.1 \times 10^8 \to 4.1 \times 10^7$). The yield cost is now visible in the data: at $n \geq 17$ and $\delta \geq 5$ the measured decomposable fraction hits zero within 16 targets (counting floor $2^{-\delta}$), so the sweep's usable range ends where the floor bites - exactly the trade the counting boundary predicts.

F10 - the arity+size lever drops two degrees. On the symmetric-S_4 m=3 encoding, shrinking the base from $l = 3$ to $l = 2$ at $n = 15$ ($\delta : +6 \to +9$) drops D^* from 5 to 3; at $n = 19, l = 3$ ($\delta = +10$) the standard and geometric systems refute at $D^* = 4$ with 15 unknowns where the prior chained evidence sits at $D^* = 5$, and point-decomposable yield is still 0.0625 per target (geometric). The subfield cell at $n = 21, l = 3$ again refutes at $D^* = 3$ with 9 unknowns. Read together with F1: closure and heavy δ are the only measured levers that move D^* by whole degrees, and heavy δ is the one that exists at $n = 131$.

Round 3 (2026-10-14): the arity+size law `D* = l + 1`

Seven more symmetric-S_4 cells (n=21 l=2, n=19 l=2, n=23 l=2/l=3, n=29 l=3/l=4; n=21 l=4 lost to a PolyBoRi segfault, preserved in the logs) plus the n=23 m=2 top dims (9, 10). $n = 29$ has $\mathrm{ord}_{29}(2) = 28 = n - 1$, 131's full-length cyclotomic order. All cells completed except the segfault.

F11 - at heavy δ, $D^* = l + 1$, independent of the field degree. Every heavy-δ symmetric-S_4 cell sits on the diagonal: $D^* = 3$ at l = 2 for n = 19, 21, 23; $D^* = 4$ at l = 3 for n = 19, 23, 29; $D^* = 5$ at l = 4 for n = 29 (and the near-determined n=15 l=3 cell is the one off-diagonal point at $D^* = 5$). The m=3 system's degree is set by the base dimension l alone - the same law F2 found for m=2 - so the per-system Macaulay cost of the oracle is $2^{O(l)}$ at any field size, including $n = 131$.

F12 - orbit-amplified yield at heavy δ. At n = 29, l = 3 (naive counting floor $|V|^3/2^{29} = 2^{-20}$) the measured point-decomposable fraction is 0.031 (2^{-5}) over 32 targets - about 2^{15} above the naive floor. The Frobenius phases and permutation orbits of the Koblitz family amplify the point-level yield far above the subfield-free counting bound; the same amplification appears at every heavy-δ cell (n = 23 l = 2: measured 2^{-5} against floor 2^{-17}). This is the quantity that decides whether a heavy-δ oracle can still harvest relations at $n = 131$, and it must be re-priced at m = 83 before any transfer claim.

Round 3 verdict for the challenge curve

The measured lever set for an ECC2K-130 index-calculus oracle now reads: (i) $D^* = l + 1$ at heavy δ - per-system cost grows only with the base dimension, never with n (F11); (ii) the per-relation δ gain grows with n (F9); (iii) orbit structure amplifies yield 2^{15} above the naive floor at heavy δ (F12); (iv) multiplicative closure, the strongest degree lever, does not exist at n = 131 (F1, F7). A drastic speedup at 131 therefore needs an m=3-or-higher oracle whose base dimension l stays small (l = 2..4 keeps $D^* \leq 5$) while orbit-amplified yield keeps relations flowing - the m=83 gate is the named, unmet validation step.

What this says about a drastic ECC2K-130 speedup

Every measured degree lever (F1, F3, F6) runs through multiplicative closure of the base, and none of the closure-free structures (windows, geometric special subspaces, weight classes) lower D^*. At $n = 131$ no proper subfield exists, so on the challenge curve the Gröbner-oracle track's per-system cost can only be trimmed by **size** levers (F4, F5: smaller effective bases, interior optimum near $n' \approx n/3$, $\delta \approx n/3$) worth 10–300x per relation at toy scale, and by **arity** levers (the m=5/6-sum and 6-sum yield models of `RESEARCH_ECC2K130_IC_FEASIBILITY.md` §4), while the Frobenius-invariant weight-class bases (F7) cut the number of systems by n with the windows verified at 131. The counting floor (yield $\approx |V|^m/2^n$) fixes how small a base can be before relations vanish; at $n = 131$ that floor is what forces $K \geq 2 \times 10^7$ columns in the 4-sum ladder, so the size levers trade against it and must be re-priced at m=83 under the repository's confidence gate before any transfer claim.

Classification and falsification record

H1 confirmed (closure lowers D^*; windows do not). H2 confirmed ($r = 0.87$ with n', interior optimum of W\/rel). H3 confirmed (6.5–11x at $\delta = +3$; the $2^{-\delta}$ yield cost is visible in the recorded fractions). H4 confirmed (geometric = standard on D^*). H5 confirmed ($r(D^*, n) = -0.04$). Chained m=3 re-measurement: not completed (engine budget, preserved failures); prior $D^* = 5$ figure stands as prior evidence. All numbers are toy-scale stage diagnostics; nothing here is an end-to-end or sub-ρ claim, and the m=83 gate (AGENTS.md §8a) is unmet by this study.